

Technology Stack

Date	25 June 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies used to build the Credit Card Approval Prediction System, including the programming languages, frameworks, deployment platform, and tools. The selected technology stack ensures the application is efficient, scalable, and easy to maintain. Technology

Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g.,HTML/CSS)	HTML5, CSS3	Used to develop a simple, responsive, and user-friendly web interface for entering applicant details and displaying prediction results.

2	Backend / Server-Side (Python Framework)	Python (Flask)	Flask is a lightweight Python framework used to connect the web application with the trained machine learning model and process prediction requests.
3	Database / Data Storage	CSV Files (Pandas)	The application uses CSV datasets (application_record.csv and credit_record.csv) for data preprocessing, training, and prediction.
4	Cloud / Hosting / Deployment	Render	Render is used to deploy the Flask application online, providing reliable hosting and easy access through a web browser.

5	Version Control & CI/CD	Git & GitHub	Git is used for version control, while GitHub stores the source code, tracks changes, and enables collaboration and deployment.
6	Machine Learning & Other Tools	Scikit-learn, Pandas, NumPy, Joblib, Matplotlib, Seaborn, Kaggle Dataset, VS Code	Scikit-learn is used for model training, Pandas and NumPy for data processing, Joblib for saving the trained model, Matplotlib and Seaborn for visualization, Kaggle provides the dataset, and VS Code is used for developmen